KARAN KHOKHAR

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING UNDERGRADUATE

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PROFESSIONAL SUMMARY

Motivated AI & Machine Learning undergraduate with solid foundations in Python, algorithms, and practical software engineering. Proven ability to build accessible, high-performance web applications, implement modular data structures, and explore neural network architectures. Committed to shipping reliable open-source code and eager to contribute to forward-thinking software engineering and machine learning teams.

EDUCATION

CT Group of Institutions, Shahpur Campus 2026 - 2030 (Expected)

Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning

• Relevant Coursework: Data Structures & Algorithms, Programming in Python & C, Discrete Mathematics, AI Foundations.

TECHNICAL SKILLS

- **Languages:** Python, C, JavaScript, SQL, HTML5, CSS3
- **AI & Machine Learning:** Neural Network Fundamentals, Model Evaluation, PyTorch (In Progress), Optimization
- **Developer Tools:** Git, GitHub, Linux, Bash, Vite, VS Code, Node.js
- **Core Computer Science:** Data Structures, Algorithm Design, Object-Oriented Programming, Modular Systems

KEY PROJECTS

Personal Developer Portfolio & Lab Platform **[GitHub Repository]**

React, Vite, Material UI, JavaScript, Responsive Web Architecture

• Designed and built an ultra-clean developer portfolio inspired by modern dark-mode software aesthetics with 100% mobile responsiveness.
• Architected "The Lab" workbench for prototyping and testing small experiments across idea, in-progress, and completed stages.
• Maintained structured, evidence-based project logs detailing technical briefs, architecture decisions, outcomes, and retrospective learnings.

Python Systems & Algorithmic Learning Repository **[GitHub Repository]**

Python, Data Structures, Algorithms, Unit Testing

• Implemented core computer science algorithms and modular data structures from scratch in Python to build rigorous intuition.
• Analyzed computational complexity trade-offs (Time & Space complexity) across search, sorting, recursion, and dynamic structures.
• Maintained a clean, version-controlled repository serving as an open public record of continuous programming progression.

CURRENT DEVELOPMENT FOCUS

• Deep-diving into neural network mathematics, tensor operations, and PyTorch internals for machine learning applications.
• Strengthening low-level programming concepts, memory layout, and systems programming fundamentals using C.
• Actively building open-source projects on GitHub with emphasis on clean commits, modular design, and reproducible testing.